

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Settleable Solids	WQP 007	1.0	

PREPARED BY			
Agency	Responsible Person	Signature	Date
Caribbean Coral Reef Institute	Dr. Martha Ricaurte Chica		
Caribbean Coral Reef Institute	Dr. Travis Courtney		5/10/23
Caribbean Coral Reef Institute	Carla L. Mejías Rivera		
Caribbean Coral Reef Institute	Manuel A. Nieves Ortiz		

VERSION HISTORY			
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1.0	5/10/23	Initial Release	TAC
1.1	9/27/23	Added 48 hour handling time to SOP to match QAPP.	TAC

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1. Title

Standard Operating Procedure for volumetric determination of Settleable Solids.

2. Applicability

This SOP applies to the determination of Settleable Solids at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater surface and bottom seawater Settleable Solids at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater Settleable Solids data following the Quality Project Plan.

4. Method

Standard Method SM 2540-F (Baird et al., 2017).

5. Health and Safety

- 5.1. This material is classified as not hazardous under OSHA regulations.

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.
- 6.3. Do not use solvents, alkaline, or abrasive cleaners or other abrasives such as steel wool or scouring powder to clean the Imhoff Cones.
- 6.4. Oil and grease may interfere with the results.
- 6.5. High mineral content may interfere with the results.

7. Training

- 7.1. Personnel shall be trained in the proper sample collection, handling and analysis procedure for Settleable Solids.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Imhoff Cone
- 8.2. Imhoff Cone Rack
- 8.3. 1 L Polypropylene Plastic Bottle (for sampling)
- 8.4. Van Dorn bottle

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9. Procedure

9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close Van Dorn at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push to open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7. Rinse the 1L plastic bottle for sample collection three times with water from the Van Dorn bottle.
- 9.1.8. Collect the sample until the bottle overflows.
 - 9.1.8.1. Refer to Appendix J for the seawater sample collection order.
- 9.1.9. Place the sample bottle inside the cooler until analysis in the lab.
 - 9.1.9.1. Sample must be analyzed within 48 hours in accordance with QAPP.

9.2. Sample Processing

- 9.2.1. Place the Imhoff cones into the corresponding cone rack.
- 9.2.2. Assess the Imhoff cones for any visible damage.
 - 9.2.2.1. In case of visible damage, refer to section 12 for further actions.
- 9.2.3. Bring the sample to room temperature to avoid obscuring the measurement by condensation of the cold sample.
- 9.2.4. Shake the sample bottle well before use.
- 9.2.5. Pour the sample into one Imhoff cone to the 1 L mark.
- 9.2.6. Let the sample settle for 45 min.
- 9.2.7. After the initial settling time, gently agitate the sample near the sides of the cone with a rod or by gently spinning.
- 9.2.8. Let the sample settle for an additional 15 minutes.
- 9.2.9. Record volume of settled solids in bottom of the cone as millimeters per liter.
 - 9.2.9.1. In case of pockets of liquid between settled particles, estimate volume of the pockets and subtract it from volume of settled solids.
 - 9.2.9.2. In case of floating material, exclude the measurement from the settled matter volume. Lowest range of settled solids measurements may fluctuate between 0.1 to 1.0 mL/L.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Sample total settleable solids (mL/L) will be recorded in the total settleable solids lab notebook.
- 10.3. Data contained in hard copy form shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.4. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.5. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. After the process is completed, clean the Imhoff cones with distilled water and a conical shape

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brush.

- 11.2. Remove bottom closure and rinse thoroughly.
- 11.3. Let the cones dry until next use with caps on.
- 11.4. Do not use strong acids, strong bases, or organic solvents with the Imhoff cones as these may damage the cone.

12. Troubleshooting

- 12.1. If the bottom closure is not screwing adequately, replace it immediately.
- 12.2. If the Imhoff cone breaks, discard it properly.

13. QA/QC

- 13.1. As per QAPP Table 4, seawater Settleable Solids should be measured in milliliters per liters (mL/L) with a precision of <20%.
- 13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

14. References

- 14.1. Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.